

What a Language Model Does with a Solver Certificate: An Answer-Evidence Decomposition Across Ten Models

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Abstract

Solver-, tool- and retrieval-augmented systems report large accuracy gains, routinely read as evidence that the model used the supplied material. Two mechanisms give the same number: reading a supplied answer, and reasoning over supplied state. The standard control, a same-shape irrelevant record, separates neither. We build one that does — holding every surface property fixed while destroying validity alone — and apply it across four sealed experiments and two robustness studies, 34,584 scored responses re-derivable from released receipts.

The decomposition is the contribution; what it returns is not. The validity share is 98.3% on one checkpoint and 40.4–51.4% on another depending on the denominator, so a single-model mechanism result should not be generalised, ours included. Two negative results replicate everywhere: given a valid-looking certificate whose fabricated final rule establishes the negation, models follow it on 189, 186 and 192 of 192 items; and a three-candidate test refutes the reading that they detect invalidity at all, the strongest substrate abstaining on 0 of 192 broken certificates while completing 76.

We also report what the design cannot support: the 2×2 is ceiling-limited, viability is a property of (model $\times$ arm $\times$ runtime) rather than of a model, and two prespecified criteria of our own were invalidated by our own data.

1 Introduction

Pairing a language model with an external solver, tool or retrieval index reliably raises accuracy on tasks the model cannot do alone. Systems that translate a problem into a formal representation, dispatch it to a symbolic engine and return the verdict through the model report large end-to-end gains across logical reasoning, program execution, API use, theorem proving and retrieval-supported generation [Pan et al., 2023, Olausson et al., 2023, Ye et al., 2023, Gao et al., 2022, Schick et al., 2023, Yang et al., 2023, Lewis et al., 2020]. Those gains are routinely read as evidence that the model *used* the supplied material.

That reading is underdetermined. Two mechanisms pro-

duce the same accuracy number. Under *answer reading*, the supplied record contains the conclusion, or something that co-occurs with it, and the model copies out the implied response. Under *state use*, the model works with the supplied premises and completes the remaining inference itself. The first says nothing about what the model does with reasoning; the second says something specific. The standard control is a same-shape but irrelevant record, which rules out proof length, generic validity and formatting. It does not rule out answer reading, because a *relevant* certificate stays answer-diagnostic even when the response token is withheld: a prefix stopping one step short still names the query’s subject and predicate in adjacent lines, so a model that only checks surface co-occurrence reproduces every number a reasoning model would.

Separating the two requires a control that holds surface form fixed and destroys validity alone. This paper builds one and applies it across four sealed experiments on frozen small models, scoring by direct next-token likelihood over verified single-token candidates so that no free text is parsed. Experiment 1 reads a five-arm certificate factorial as a 2×2 of *answer evidence* $\times$ *proof state*. Experiment 2 holds every surface property fixed across ten arms while breaking only the logic, and varies entity repetition, line order and truncation depth alongside a valid-looking certificate whose fabricated final rule points the wrong way. Experiment 3 scores ten models on a three-class panel whose *undetermined* items are certified by forward-closure saturation. Experiment 4 rescues the broken-chain items with a third candidate, separating detection of an invalid certificate from failure to complete one.

The method is the contribution; what it returns is model-specific. Breaking validity alone removes almost the whole state effect on one checkpoint — a validity share of 98.3% — and 40.4% on another, which answers 58 of 96 broken certificates correctly where the first manages 1. We offer the decomposition and decline to generalise the value it returns, ours included. Three results do hold everywhere: the inference is one step deep and requires the final rule to *follow* the premise it fires on; it is *not* detection, since a broken certificate *lowers* abstention; and none of the three checkpoints resists a corrupted apparatus, answering incorrectly on 189, 186 and 192 of 192 items given a

certificate whose fabricated final rule establishes the negation.

Two measurement points shape how this may be reported. Accuracy alone is not interpretable: one arm scored 50.0% on a balanced panel at $d' = 0.00$, so sensitivity and criterion accompany it throughout [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], with the exact McNemar test [McNemar, 1947] and BCa intervals [Efron, 1987]. And the claim boundary is strict: on the Qwen checkpoints every arm without a near-complete chain returns $d' = 0$, so those models do not reason unaided and high accuracy with full solver material describes a faithful transcriber. Gemma-3-4b is not in that position (22/96 unaided), which is why its share is reported against two baselines. System-level effects are never upgraded into claims about the model’s own reasoning.

The contributions are: (i) an *answer-evidence decomposition* separating solver-supplied answers from solver-supplied reasoning state, reusable for any tool-augmented system; (ii) a *surface-matched invalidity control* isolating validity from every co-occurring surface cue, certified per item; (iii) a *forward-closure certifier* for underdetermined items, validated at 23,240/23,240 against an independently labelled corpus [Tafjord et al., 2020] and against nine adversarial classes; (iv) evidence that apparent use of proof state can be surface exploitation, that the behaviour is completion rather than detection, and that no checkpoint resists a corrupted certificate; and (v) a ten-model sweep showing viability to be a property of (model $\times$ arm), with proof state *degrading* three of ten interfaces [Team et al., 2024b, Grattafiori et al., 2024, Team et al., 2025, Microsoft et al., 2025, Team et al., 2024a, Allal et al., 2025].

2 Related Work

2.1 Faithfulness of displayed reasoning

Chain-of-thought prompting made intermediate steps part of the output [Wei et al., 2022], and a body of work then asked whether the answer rests on them. Turpin et al. [2023] show explanations systematically omit biasing features that demonstrably drive the prediction. Lanham et al. [2023] intervene on the trace — truncating it, and injecting mistakes — and read the change in the answer as a measure of conditioning; they find wide variation and that mistakes are often ignored. The same manipulations have been applied to naturally worded prompts [Arcuschin et al., 2025] and to models trained for long traces, where answer-determining hints are verbalised only a minority of the time [Chen et al., 2025]; Lyu et al. [2023] respond architecturally, making the trace a program executed by a solver.

The limitation is one of object rather than method. This literature intervenes on *model-generated* traces, asking

whether the model’s own account is honest. It does not address *externally supplied, formally verified* state, where the question inverts: does the model track a derivation it did not produce? We adapt truncation and mistake injection to solver certificates and add a control this literature does not use — an invalid certificate matched to a valid one on every surface property, certified per item not to settle the query. That also changes what a null means: ignoring an injected mistake in one’s own trace is unfaithful, whereas following a corrupted external certificate is unsafe.

2.2 Solver-augmented and tool-augmented reasoning

Neurosymbolic pipelines put the formal step outside the model: translating into a symbolic language and repairing from solver errors [Pan et al., 2023], parsing into first-order logic for a prover [Olausson et al., 2023], dispatching a declarative specification [Ye et al., 2023], or emitting a program for an interpreter [Gao et al., 2022]. The pattern extends to self-supervised API use [Schick et al., 2023], premise selection in interactive theorem proving [Yang et al., 2023], and retrieval as general external evidence [Lewis et al., 2020].

All report end-to-end accuracy and attribute the gain to the pipeline, leaving two things open. The *reasoning state*’s contribution is not separated from the *answer*’s, since the solver’s output contains both. And the solver is assumed correct, so behaviour under wrong-but-well-formed output is unmeasured — though the model side is demonstrably not neutral to context, with irrelevant material degrading accuracy [Shi et al., 2023] and coherent evidence overriding parametric knowledge [Xie et al., 2023]. We decompose the solver’s contribution into an answer channel and a state channel, and add an arm whose certificate is valid-looking but points the wrong way.

2.3 Synthetic deductive reasoning benchmarks

The task family descends from rule-based deductive datasets in natural language: RuleTaker [Clark et al., 2020], ProofWriter, which added generated proofs and open-world semantics with an explicit unknown class [Tafjord et al., 2020], PrOntoQA, whose chains parse into symbolic proofs and whose individual steps are usually correct while proof planning is not [Saparov and He, 2022], and FOLIO’s human-written premises with verified annotations [Han et al., 2022]. Two properties matter here: order effects are large in deductive prompting [Chen et al., 2024], and these sets are public and predate the models evaluated on them, so contamination cannot be excluded [Deng et al., 2024, Golchin and Surdeanu, 2023, Wu et al., 2023].

They supply items and proofs but not the matched adversarial certificate variants an attribution question needs. We generate items after model release with per-item nonce vocabularies and construct the matched invalid, misleading, order-scrambled and depth-truncated certificates the public sets do not provide. A public open-world corpus is used only to check the forward-closure certifier; no item from it is scored by any model here.

2.4 Response bias and the limits of accuracy

Prompted classification is biased rather than merely noisy. Few-shot accuracy swings from near chance to near state of the art with format and example order, because the model carries a prior toward particular answer strings [Zhao et al., 2021]; option order alone moves multiple-choice accuracy substantially [Pezeshkpour and Hruschka, 2023], tracing to a token-level prior over option identifiers rather than option content [Zheng et al., 2023]; and likelihood scoring adds a confound, since strings compete for probability mass independently of correctness [Holtzman et al., 2021]. On abstention, models can partially assess whether their own answers are right [Kadavath et al., 2022] and be evaluated on recognising questions that admit no answer [Yin et al., 2023].

Evaluation practice in solver-augmented reasoning has not absorbed this: bare accuracy remains the reported quantity. On a balanced panel a degenerate single-label responder is indistinguishable from a chance-level competent one, and both from a partial discriminator with a shifted criterion. Signal detection theory separates exactly these [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], and this paper therefore reports d' and criterion c alongside accuracy for every arm, with paired significance from the exact McNemar test [McNemar, 1947] and interval estimates from the bias-corrected accelerated bootstrap [Efron, 1987]. Two concrete cases motivate the practice rather than illustrate it: a baseline arm scoring 50.0% on a balanced panel turns out to have $d' = 0.00$, and a prespecified qualification criterion based on a single class recall was passed by models that answer one label almost everywhere. Both were caught only because sensitivity was reported separately from accuracy; one of the two is a criterion of this paper’s own, replaced post hoc and labelled as such.

3 Method

3.1 The answer-evidence decomposition

A solver certificate carries two things at once: the *answer* to the query, and the *state* from which that answer follows. End-to-end accuracy conflates them. We separate them by arranging arms as a 2×2 of answer evidence by proof

state, with a no-material baseline outside the square.

Let IRRELEVANT supply a valid derivation for a different entity (no answer, no relevant state), CONCLUSION_ONLY the derived terminal literal alone (answer, no state), PROOF_PREFIX the chain with that literal withheld (state, no answer), and FULL both. A contrast along an *edge* of this square moves one factor; a contrast along the *diagonal* moves two, and a null on it attributes to neither. This is not a statistical subtlety but an identification one: the prespecified primary contrast of the experiment we re-analyse, PROOF_PREFIX minus CONCLUSION_ONLY, is exactly that diagonal.

Reading the four edges instead gives a state main effect, the mean of the two edges along which state is added, and an answer main effect defined symmetrically.

3.2 Surface-matched invalidity

A prefix that stops one step short is not a clean state manipulation. It still names the query’s subject and predicate in adjacent lines, so surface co-occurrence remains available. To isolate validity we construct BROKEN_CHAIN: one intermediate link is cut so the displayed lines no longer reach the query, while every surface property is held fixed.

Concretely, against the matched TRUNCATE_1 arm we preserve the query-entity occurrence count (audited per item, not approximately), the presence of the query predicate in the final rule, the line count, the exact fact/rule/literal type sequence, and token length within a recorded tolerance. Every *rule* displayed is a real rule of the theory; exactly one derived literal is false. All 192 items are certified by forward-closure saturation (section 3.3) to leave the query *undetermined* under the displayed lines. The decomposition follows:

$$\text{total state effect} = \text{TRUNCATE_1} - \text{IRRELEVANT}, \quad (1)$$

$$\text{surface component} = \text{BROKEN_CHAIN} - \text{IRRELEVANT}, \quad (2)$$

$$\text{validity component} = \text{TRUNCATE_1} - \text{BROKEN_CHAIN}. \quad (3)$$

Four companion arms separate the remaining confounds. SAME_ENTITY_IRRELEVANT supplies a valid derivation about the *query* entity toward an unrelated predicate: it names the query subject four times where IRRELEVANT names it zero times, and neither contains the query predicate, so the pair isolates entity repetition alone. SHUFFLED presents TRUNCATE_1’s lines in seeded random order, separating a chain from a bag of statements. TRUNCATE_2 and TRUNCATE_3 withhold two and three steps, giving a depth ladder. MISLEADING supplies a valid-looking chain whose single fabricated final rule establishes the query’s negation; the ground-truth answer is unchanged,

so a model that follows the supplied state answers incorrectly.

3.3 Certifying underdetermination

Showing that a query *is* derivable is easy: exhibit the proof. Showing it is *not* requires saturating the theory’s forward closure. Our certifier operates under open-world semantics with explicit negation, so a negative literal is derivable only if something derives it, never by failure to prove the positive. Rule antecedents are ordinary literals, with no negation as failure, which makes naive forward chaining sound and complete on this fragment.

Two safeguards matter for generated theories. The chaining loop carries an explicit round cap and reports a *saturated* flag, so a malformed theory fails loudly rather than spinning; and any theory deriving both an atom and its negation is marked inconsistent and its items rejected rather than labelled. An item is *undetermined* when neither the query nor its negation appears in the closure.

Validation has three layers, because a single one would not be worth much. Against the open-world splits of a public corpus [Tafjord et al., 2020] at depths 0–3 and 5, 23,240 of 23,240 questions agree, including 10,440 undetermined. That is weaker than it sounds. Those labels also come from forward chaining over the same fragment, so the agreement largely measures one chainer against another, and it covers none of the generated nonce theories. We therefore add two checks with different failure modes. Nine adversarial theories probe cycles, chains past the round cap, derived negation, absent symbols and entity-specific rules; two of the nine must *fail* rather than answer, since an inconsistent theory must be rejected and a cap overrun must report failure rather than a wrong verdict. All nine pass, and the cases are released. And a *differential* test on the generated panels — an independently written reference using naive fixpoint over the exhaustive Herbrand base, a deliberately different algorithm — agrees with the certifier and the sealed authority on 192/192 items. That reference encodes the same intended semantics, so it catches implementation error rather than a misconception about the semantics; we claim the former only. No item enters a sealed panel without a certification record.

3.4 Measurement: sensitivity, not accuracy

Accuracy on a balanced panel is uninterpretable when the responder is biased. One arm below scores 50.0% — apparent chance — with an identical 9/96 “Yes” rate in *both* classes, that is $d' = 0.00$: no discriminative signal whatever, at or below the trivial always-one-label strategy. We therefore report sensitivity d' and criterion c alongside accuracy for every arm [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], using the loglinear correc-

tion for extreme rates. For the three-class panel the headline scalar is balanced accuracy, the mean of per-class recalls, under which chance is 33.3% and a single-label responder scores exactly chance.

Paired contrasts report the complete 2×2 table, the exact two-sided McNemar test [McNemar, 1947] as the decision criterion, and a seeded 100,000-resample paired bias-corrected accelerated bootstrap interval [Efron, 1987], with Holm adjustment within declared families.

4 Experimental Setup

4.1 Task, panels and scoring

All four experiments use rule-chaining queries over small theories of facts and universally quantified rules. Table 4 lists the panels. The sealed panels are procedurally generated after model release with per-item nonce vocabularies, are mutually disjoint, and fix derivation depth at four by construction, recorded per item. The public corpus is used *only* to validate the certifier; no item from it is scored by any model here.

Experiment 4 generates no panel: it rescores Experiment 2’s items with a third candidate available, so failing to complete a broken certificate can be told apart from recognising that it is broken. Theories, certificates and queries are byte-identical; only the instruction line and candidate set change, and the change is recorded in the seal. Its NONE arm withholds the solver record entirely under that same instruction, fixing the abstention floor independently of whether a record is present.

Responses are scored by direct next-token likelihood over verified single-token candidates: Yes/No for Experiments 1 and 2, Yes/No/Unknown for Experiments 3 and 4. The model generates no free text, so nothing is parsed and no verdict is inferred from prose. Every prompt is four lines — instruction, theory, solver record, query — of which only the solver record varies across arms; the theory is visible in all arms.

4.2 Models

Experiments 1 and 2 use the same frozen 3B instruction-tuned checkpoint at 4-bit precision [Team et al., 2024b], pinned to byte-identical weights, so the anchors of the first transfer to the second without a weights caveat. Experiment 3 scores ten instruction-tuned models from 0.5B to 7.6B [Team et al., 2024b, Grattafiori et al., 2024, Team et al., 2025, Microsoft et al., 2025, Team et al., 2024a, Allal et al., 2025], all at bfloat16 so precision is not confounded with model identity. An eligibility screen confirmed that all ten load, complete a forward pass, and tokenise all three candidates as single distinct tokens.

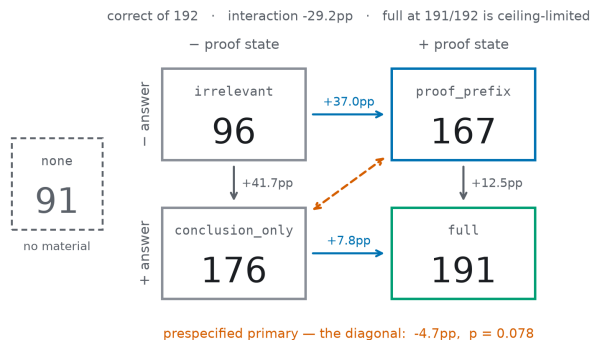


Figure 1: The five arms form a 2×2 of answer evidence by proof state, with the no-material baseline outside it. Each cell gives correct items of 192. Arrows mark the four one-factor edges; the dashed diagonal is the prespecified primary, which moves both factors at once and therefore attributes its null to neither.

4.3 Sealed protocol

Panels are hash-stamped before scoring and the runner refuses a modified panel. Scoring takes the highest-likelihood candidate token; on an exact tie the first candidate in the declared order wins, which across Experiment 2’s 5,760 responses occurs twice. Each response is written to a receipt carrying a SHA-256 of its own contents, and the answer authority is opened only once every receipt exists, so nothing can be tuned to a result. Audits depend on the standard library alone, so verification never requires the environment to resolve. In total 34,584 responses were scored across 34 model-runs, with zero receipt digest failures.

Experiment 3 makes a deliberate protocol change: *no gate blocks the run*. Where earlier practice excluded a model that failed a delivery screen, delivery is itself the capability under test here, so a model that cannot emit a verdict reliably is a finding about substrate viability rather than a reason to drop it. Tooling failures — a model that will not load, or whose tokeniser cannot represent the candidates — remain exclusions and are not findings; the screen found none.

5 Results

5.1 Reading the factorial as a 2×2

Figure 1 shows the five arms of Experiment 1 arranged by section 3.1. The prespecified primary contrast, `PROOF_PREFIX` – `CONCLUSION_ONLY`, was -4.7pp with exact McNemar $p = 0.078$ and was reported as a null. It is the diagonal.

The four edges are reported in table 5. **This reading is post hoc:** a different analysis of sealed data, not a re-run. Every value is re-derived from the 960 raw receipts by a

script that reconstructs the answer key and asserts it reproduces all five published per-arm counts before reporting anything. The state main effect is $+22.4\text{pp}$ and the answer main effect $+27.1\text{pp}$.

The **interaction** is -29.2pp , larger than either main effect. The design is also **ceiling-limited**: `FULL` reaches 191/192 (99.5%), leaving only 13.0pp of headroom above `PROOF_PREFIX`. Both “other factor present” edges are therefore compressed, and the averaged main effects deflated with them.

This bears on one tempting reading. The answer literal appears to retain 30% of its value once state is present (12.50/41.67), which would suggest the two channels are partly redundant. But with `FULL` at ceiling that ratio largely measures the headroom left, not the redundancy, so we do not draw the conclusion. Separating them needs a design with `FULL` away from ceiling.

Splitting by class overturns the baseline description. The model answers `No` to 88% of items. Table 6 gives the signal detection summary: `IRRELEVANT` returns an identical 9/96 “Yes” rate in both classes, $d' = 0.00$. A trivial always-`No` strategy scores 96/192; `NONE` scored 91 and `IRRELEVANT` 96, at or below it. The contradicted class sits at ceiling in every certificate arm and contributed 0 of the 21 discordant pairs in the primary contrast, so the effective sample was 96, not 192.

5.2 Validity versus surface overlap

Experiment 2 holds surface form fixed and destroys validity alone. The primary contrast is `TRUNCATE_1` – `BROKEN_CHAIN` = $+60.4\text{pp}$: exact McNemar $p = 6.9 \times 10^{-18}$, BCa $[+49.0, +68.8]$, from a paired table of 1 both correct, 58 `TRUNCATE_1`-only, 0 `BROKEN_CHAIN`-only and 37 neither. It is evaluated on the entailed subset ($n = 96$), as declared before the run, because the contradicted class contributes no discordant pairs. Sensitivity moves with it: d' drops by $+2.446$, satisfying the same-sign requirement declared in advance, so this is a change in discrimination and not in criterion placement.

The decomposition is therefore a total state effect of $+61.5\text{pp}$ comprising a surface component of $+1.0\text{pp}$ ($p = 1$, 95% BCa $[+0.0, +3.1]\text{pp}$) and a validity component of $+60.4\text{pp}$: a **validity share of 98.3%**, **95% BCa** $[94.4, 100.0]$. The interval bootstraps the whole ratio over paired item resamples rather than its numerator alone. Figure 2 and table 1 give the full arm set. Section 5.4 shows this share is specific to this checkpoint and does not generalise.

Three supporting contrasts sharpen the reading. Entity repetition explains nothing: `SAME_ENTITY_IRRELEVANT` – `IRRELEVANT` = $+0.0\text{pp}$ ($p = 1$), despite the former naming the query subject four times and the latter not at all. Order matters *on*

Table 1: Experiment 2: ten matched arms scored on three checkpoints, entailed subset (correct of 96, with signed d'). Only the solver record differs between arms. The share rows give the validity component as a fraction of the total state effect, against the IRRELEVANT control and against the unaided NONE arm; the two coincide wherever unaided competence is zero.

Arm	Qwen2.5-3B 4-bit		Qwen2.5-3B bf16		Gemma-3-4B	
	/96	d'	/96	d'	/96	d'
NONE	0	0.00	0	0.00	22	1.83
IRRELEVANT	0	0.00	0	0.00	2	0.62
SAME_ENTITY_IRRELEVANT	0	0.00	0	0.00	3	0.77
TRUNCATE_3	0	0.00	0	0.00	4	0.88
TRUNCATE_2	0	0.00	0	0.00	4	0.88
TRUNCATE_1	59	2.85	72	3.23	96	5.13
BROKEN_CHAIN	1	0.41	1	0.41	58	2.83
SHUFFLED	14	1.53	12	1.43	86	3.80
MISLEADING	0	-4.36	0	-4.06	0	-5.13
FULL	96	5.13	96	5.13	96	5.13
Validity share vs IRRELEVANT	98.3%		98.6%		40.4%	
Validity share vs NONE	98.3%		98.6%		51.4%	

this checkpoint: $\text{TRUNCATE}_1 - \text{SHUFFLED} = +46.9\text{pp}$ ($p = 6.8 \times 10^{-13}$, Holm 1.4×10^{-12}), so it is not reading a bag of statements — though section 5.4 shows this too is checkpoint-specific, and section 5.3 shows *order* is the wrong word for it. And depth is a cliff, not a slope: $\text{TRUNCATE}_2 - \text{TRUNCATE}_1 = -61.5\text{pp}$ ($p = 3.5 \times 10^{-18}$, Holm 1.0×10^{-17}), with TRUNCATE_2 scoring identically to supplying no certificate at all. Holm adjustment is within the secondary family declared in section 3.4; the four post hoc edges of table 5 carry Holm values 6.6×10^{-24} , 4.5×10^{-19} , 2.4×10^{-7} and 6.1×10^{-5} . Five arms are indistinguishable — 96/192, 0/96 entailed, $d' = 0.000$, $c = 2.565$ — so behaviour is binary: either the supplied chain reaches one step from the answer, or the model is blind to it.

5.3 Not adjacency, but direction

SHUFFLED is one seeded permutation per item, and scrambling moves three things at once: the gap between the final rule and the premise it fires on, the order of that pair, and everything else. We re-scored the same items under eight permutations each on all three checkpoints, recording where those two lines landed. An identity arm reproduces TRUNCATE_1 's bytes and returns its receipts 192/192 throughout, and the published +46.9pp falls inside the resulting range on every checkpoint.

Distance does nothing; order does everything. Gap 1 and gap 7 are indistinguishable (18.1% against 24.0% on the 4-bit checkpoint), so separating the two lines does not hurt. Reversing them does: rule *after* its premise gives 26.8%, 22.4% and 94.9% across the three checkpoints, rule *before* gives 5.8%, 4.8% and 84.9%; paired within item on the 4-bit checkpoint, 45 items favour rule-after against 2 ($p = 1.6 \times 10^{-11}$). The requirement is therefore not adjacency — section 6.1 said so and was wrong

— but that the final rule *follow* its premise. Canonical order is better again (61.5% against $\sim 27\%$), so the chain before the last step matters too.

5.4 The validity share does not generalise

The decomposition of section 3.2 was run on two further checkpoints: the same model at bfloat16 rather than 4-bit, which isolates quantisation from model identity, and a different family. Table 1 carries all three; fig. 3 plots the split.

Quantisation is the smaller effect: bfloat16 raises TRUNCATE_1 from 59 to 72 of 96 but leaves the share essentially unchanged, so precision moves the magnitude of the state effect without moving its composition. Model identity does not behave that way. Gemma-3-4B answers BROKEN_CHAIN correctly on **58 of 96** items where the other two manage 1, so the same control that isolates inference on one checkpoint exposes substantial surface exploitation on another.

Which baseline. The share divides by the effect against IRRELEVANT, which on the Qwen checkpoints also scores 0/96, so the choice is invisible there. Gemma-3-4B answers 22 of 96 entailed items with no certificate at all — 20 of those 22 also correct under BROKEN_CHAIN — so against NONE its share is 51.4%, not 40.4%. Both are in table 1, and we make no claim that most of its effect is surface overlap: on the conservative denominator the split is close to even. The baselines diverge exactly when a substrate has unaided competence, and that divergence is the finding rather than either number.

Order-sensitivity fails to replicate alongside the share: SHUFFLED scores 14 and 12 of 96 on the Qwen checkpoints and 86 on Gemma-3-4B. Reading a certificate as a bag of statements is a checkpoint-level property too.

We therefore do not claim that solver-supplied state is used as inference in general. On the checkpoint of sec-

tion 3.2 it overwhelmingly is; on the strongest substrate in our sweep, between half and three-fifths of the effect survives the control, depending on the baseline. What generalises is the *method* — the control separates the two wherever it is applied — not the value it returns.

One result does replicate without exception, and it is the negative one. The MISLEADING arm scores 3, 6 and 0 of 192 across the three checkpoints, with d' of -4.36 , -4.06 and -5.13 . The model that exploits surface overlap most is also the one most completely misled.

5.5 Following a corrupted apparatus

The MISLEADING arm supplies a chain identical to FULL except that its final rule, which is fabricated and absent from the theory, flips polarity. The ground-truth answer is unchanged. The model scores 3 of 192 against FULL’s 192, a paired difference of -98.4pp on all items ($p = 2.5 \times 10^{-29}$), with $d' = -4.363$.

The class split is the sharper evidence. This model’s standing prior is NO — 88% of items under NONE, at $c = 2.565$. Under MISLEADING it answers YES on **93 of the 96 contradicted items**, inverting a near-total prior because a fabricated rule told it to. Corrupted solver output does not merely fail to help; it overrides the model’s own strong default.

The model therefore follows a valid-looking certificate that points the wrong way on 189 of 192 items. Read against the 98.3% validity share, the two results are jointly informative: whatever the model is doing with a supplied derivation is sensitive to whether that derivation reaches the query, and it supplies no defence whatever when the derivation reaches the wrong conclusion. Section 5.6 shows that sensitivity is completion rather than verification — so the certificate is followed, not checked.

5.6 The model does not detect invalidity

A natural reading of section 3.2 is that the model detects a broken chain. The binary design cannot support that reading, because the correct answer under a broken certificate is the model’s default label: “detected the break” and “found no pattern to complete” predict the same response.

We separated them. The BROKEN_CHAIN items, certified undetermined under their displayed lines, were rescored with three candidates rather than two — byte-identical but for the instruction line and candidate set. A validity tracker abstains *more* when the chain is broken; a pattern completer falls back to its default, which IRRELEVANT and NONE fix from either side. Table 2 and fig. 4 give all three checkpoints on all four arms.

Abstention falls on every checkpoint against *both* references, so the direction does not turn on the baseline. The magnitude does: an irrelevant record raises abstention

34.9pp over supplying nothing at bfloat16 while the two coincide on the 4-bit checkpoint, so a Δ against IRRELEVANT is not comparable across checkpoints.

On Gemma-3-4B the delta carries no weight, and we do not rest the claim on it: it abstains on 1 of 192 items given no record at all, so -0.5pp is a no-power null. What it does instead no baseline can move — given a broken certificate it abstains 0 **times in** 192 and *completes* 76 chains that do not connect. A model detecting invalidity does not do that.

The 4-bit checkpoint shows the same behaviour: NO is 110 of 192 under **both** BROKEN_CHAIN and TRUNCATE_1, the arms differing only in YES, 1 against 61. Power is a gradient, not a switch — a checkpoint falls only as far as its floor allows, and the three had 79.7, 21.4 and 0.5pp available, using 47%, 17% and all of it. The evidence weakens in that order and points one way through-out.

Detection is refuted, not merely unsupported, and unlike the validity share it is refuted on every checkpoint, including the one exploiting surface overlap most. Its failure on BROKEN_CHAIN is a failure to complete, not a detection of invalidity — consistent with section 5.5, since a model that verified validity would reject a certificate whose final rule is fabricated and absent from the theory, and it does not.

5.7 Which models can serve as the interface

Experiment 3 scores ten models on the three-class panel in all five arms. We declared in advance that a model relays non-determination at $\geq 80\%$ recall on undetermined items with full solver material. **Eight of ten passed, and the criterion was invalid:** single-class recall is maximised by answering Unknown to everything, and four models do close to that, emitting it on 62–96% of items with three never emitting NO — table 6’s failure on a different label. We replace it **post hoc** with a floor on every class, minimum per-class recall ≥ 0.50 , retaining both verdicts in the released results. Table 3 reports all five arms for all ten models, with the per-arm verdict; fig. 5 plots the same data ordered by parameter count.

Viability is a property of (model $\times$ arm), not of a model. The decisive row is Qwen2.5-7B: minimum per-class recall 0.141 under FULL against 0.938 under CONCLUSION_ONLY, where it is second best in the sweep at 97.9%. Its contradicted-class recall collapses *when the proof state is supplied*: not an interface that cannot express three outcomes, but one the proof state breaks. **Supplying proof state degrades three of ten models**, two of them otherwise among the strongest substrates.

This bounds a claim we would otherwise have made. There is a floor near 3B *for tolerating full certificates*, since below 1.7B no arm clears it, but that is not a floor

Table 2: Three-candidate rescoring, all three checkpoints: Unknown responses out of 192 per arm, against both abstention baselines. Detection predicts a *rise* under BROKEN_CHAIN; no checkpoint rises against either reference.

Checkpoint	NONE	IRREL.	BROKEN	Δ_{NONE}	$\Delta_{\text{IRREL.}}$	TRUNC_1
Qwen2.5-3B 4-bit	153 (79.7%)	144 (75.0%)	81 (42.2%)	−37.5	−32.8	21
Qwen2.5-3B bf16	41 (21.4%)	108 (56.2%)	34 (17.7%)	−3.6	−38.5	12
Gemma-3-4B	1 (0.5%)	15 (7.8%)	0 (0.0%)	−0.5	−7.8	0

Table 3: All five arms, all ten models. Balanced accuracy (%), chance = 33.3. Viability is reported per arm, because it is not a property of the model.

Model	B	NONE	IRREL	CONCL	PREFIX	FULL	Viable in
Qwen2.5-0.5B	0.5	35.4	34.9	41.1	43.2	63.5	—
OLMo-2-1B	1.0	35.9	33.3	53.6	34.9	42.2	—
Llama-3.2-1B	1.2	33.9	33.9	57.8	36.5	37.0	—
Qwen2.5-1.5B	1.5	33.3	33.3	43.2	34.9	37.0	—
SmolLM2-1.7B	1.7	33.3	33.3	45.3	39.6	69.3	—
Qwen2.5-3B	3.0	41.1	39.1	96.4	74.0	90.6	concl, prefix, full
Llama-3.2-3B	3.2	45.8	43.2	67.2	55.2	80.7	—
Phi-4-mini	3.8	44.8	39.6	68.2	71.9	90.6	full
Gemma-3-4B	4.3	63.5	51.0	100.0	84.4	96.4	concl, prefix, full
Qwen2.5-7B	7.6	33.3	33.3	97.9	67.2	71.4	concl only

for serving as an interface and the effect is not monotonic in scale. Model is not a randomised factor, so these comparisons are descriptive.

Nor is that an artefact of where the sweep stopped. Extending the Qwen2.5 rows to 14.7B — family, tokenizer and recipe fixed, only scale varying — the harm runs 0.000, 0.000, −0.219, −0.688, +0.062 from 0.5 to 14.7B: it peaks in a mid-range band and is *absent* at the top. Proof state is not more dangerous for larger interfaces; on this task family it stops mattering.

5.8 The same 2×2 across ten models

Experiment 3’s five arms are the 2×2 of section 3.1 plus a baseline, on ten models, from receipts already collected. The mean state main effect is +8.7pp against a mean answer main effect of +21.6pp, with a mean interaction of −15.9pp; CONCLUSION_ONLY exceeds PROOF_PREFIX in **8 of 10** models and equals or exceeds FULL in 6 of 10. The answer channel dominates the state channel across the model set — a more general form of the pattern table 5 shows on one model — and Experiment 1’s state main effect of +22.4pp is not typical of it.

That mean averages over four responders whose minimum per-class recall is 0.000 in every arm, three with negative state effects; excluding them it is +14.7pp against +28.5pp for the answer channel. So the cross-model state effect is not less than half the single-model estimate, and we do not claim it is. The ordering survives either subset — answer above state, and in 8 of 10 models individually.

6 Limitations

6.1 Scope of the mechanism claim

The inference is *exactly one step deep*: TRUNCATE_2 scores identically to supplying no certificate, so the model completes a final inference when both premises are present *and the rule follows the premise it fires on*, and cannot chain two steps. Section 5.3 establishes that ordering requirement directly and rules out the adjacency reading we first gave it. Nor does it *detect* invalidity, which section 5.6 tests directly and refutes. Section 5.4 bounds the claim further: against the IRRELEVANT control the share attributable to inference ranges from 40.4% to 98.6%, and from 51.4% to 98.6% against the unaided baseline.

The full theory is visible in every arm, so the query stays derivable whatever the certificate says, and breaking a certificate is diagnostic only where the model scores 0/96 from the theory alone — true of both Qwen checkpoints and *not* of Gemma-3-4B (22/96, $d' = 1.83$). A model using a broken certificate as a pointer back into the theory would be doing inference of a different kind; on Gemma-3-4B that is a live alternative rather than a hypothetical one, since 20 of its 22 unaided successes recur under BROKEN_CHAIN. We record it as an interpretation caveat, not a controlled alternative. Separately, TRUNCATE_2 and TRUNCATE_3 necessarily drop the query predicate, which lives in the final rule, so the depth ladder confounds depth with predicate presence and must be read against SAME_ENTITY_IRRELEVANT.

6.2 Post hoc analyses and superseded criteria

Two analyses are post hoc and labelled where used: the 2×2 reading of Experiment 1 is a different analysis of sealed data, not a re-run, and Experiment 3’s viability criterion replaced a prespecified threshold the data showed to be invalid (the original is retained in the released results).

Experiment 1 also carries three design errors we report rather than repair. It prespecified a 15pp primary effect while its comparison arm reached 91.7%, leaving 8.3pp of headroom, so the target was arithmetically unreachable; it recorded no proof depth, so cannot stratify its failures; and its task identifiers encoded the semantic class, making its answer authority reconstructible from the public panel. Experiments 2–4 fix all three.

6.3 Generalisation

This is one task family, small models only, and synthetic panels. Nonce vocabularies establish item novelty, not interdependence from all relevant pretraining patterns [Golchin and Surdeanu, 2023, Deng et al., 2024].

Inference runtime. Re-scoring the sealed panel through a second stack (`transformers` on CUDA rather than `mlx-lm`), nine of ten models agree on 95.5–99.7% of responses with no verdict moving — but Phi-4-mini’s FULL arm is viable under one and not the other (0.719 against 0.203 minimum per-class recall), reproduced on two GPUs and two library versions. Viability is a property of (model $\times$ arm $\times$ runtime), and one weight mirror will not load under the second stack at all.

Prompt format. Experiment 4 changes only the instruction line and the candidate set, on byte-identical theories and certificates, and the response distribution moves substantially — a format-sensitivity result as much as a detection one, consistent with the effects this literature documents [Zhao et al., 2021, Zheng et al., 2023]. Arms are compared against a baseline scored under the same instruction, so the contrast is internally valid, but the absolute rates are not format-independent.

Permutation sampling. SHUFFLED uses one seeded permutation per item; section 5.3 adds eight per item on three checkpoints, and the published value falls inside the resulting range each time. Eight is still a sample of $8!$ orderings, and the gap-versus-direction split is read off permutations that happened to land in each cell rather than from a design that fixed them.

Quantisation. Experiments 1 and 2 use a 4-bit checkpoint and Experiment 3 bfloat16, so they describe different artefacts of the same model, and 4-bit quantisation moves exactly what we measure — next-token margins and criterion placement. Section 5.4 bounds it: bfloat16 raises TRUNCATE_1 from 59 to 72 of 96 while leaving the

share essentially unchanged, so precision affects magnitude more than composition. We nonetheless do not treat the two as one artefact.

Prior report. Experiment 1’s per-arm counts and pre-specified primary contrast appeared in an earlier unpublished report by the same authors, cited anonymously; the 2×2 reading, the response-bias finding and Experiments 2–4 are new. That report concluded the 3B checkpoint failed an open-world unknown gate whereas Experiment 3 finds the same family viable; panels, candidate sets and quantisation all differ, which plausibly explains it, but we flag the discrepancy rather than leave it to a reader who finds both.

An undetermined PROOF_PREFIX cannot contain the query predicate, since the solver has nothing to say about it. Undetermined items therefore offer fewer surface cues, and a surface-matching model will look worse on them for reasons unrelated to abstention; per-arm entity-mention counts are recorded so this is analysable.

Finally, every theory here arrives *already formalised*. Nothing tests whether a model can turn a framework described in prose into the solver’s input, the step any claim of working in an unfamiliar framework depends on — and the corruption result makes that gap consequential rather than merely open: a formalisation error would not be caught downstream, but followed.

7 Conclusion

Solver assistance improves a small model’s accuracy on rule-chaining queries, and end-to-end accuracy cannot say why. The decomposition can: arrange the arms so each contrast moves one factor, and hold every surface property fixed while breaking validity alone.

What it returns is model-specific: 98.3% of the state effect survives the control on one checkpoint and 40.4–51.4% on another, so we offer the method as the contribution and decline to generalise its value, ours included.

Three findings hold on every checkpoint, and all are negative. The inference is one step deep, and requires the rule to follow its premise. It is not detection: the strongest substrate abstains on 0 of 192 broken certificates and completes 76. And nothing defends against a wrong apparatus — given a certificate whose fabricated final rule establishes the negation, the three checkpoints answer incorrectly on 189, 186 and 192 of 192. The interface inherits the apparatus’s errors in full.

Two criteria we had prespecified failed here — an arm at 50.0% accuracy was degenerate rather than chance-level, and a viability threshold was maximised by answering Unknown to everything — and only reporting sensitivity alongside accuracy caught either. We recommend it, with the decomposition, as standard.

Reproducibility Statement

Code and data. The supplementary material accompanying this submission is self-contained: sealed panels and answer authorities, all 34,584 per-response receipts, the generators, runners, audits and analysis scripts, and the forward-closure certifier with its validation against the public corpus. Every number in this paper can be re-derived from a fresh unpack with no network access, no model weights and no external packages — the audits use the Python standard library only, so verification does not depend on resolving an environment. Panels are hash-stamped and the runners refuse a modified panel; each receipt carries a SHA-256 of its own contents, and all 34,584 verified with zero failures. The count is not typed into the manuscript: it is produced by `scripts/census_receipts_v1.py` from the released receipts and substituted at compose time.

Checking one number takes a minute. `scripts/verify.py` re-derives each headline value from the receipts and the sealed authority — not from a stored analysis file — and prints it beside the value this paper states: `python3 scripts/verify.py validity-share` recomputes the 98.3% of section 3.2 and shows the four counts it came from. `-all` does the same for the validity share, Gemma’s two baselines, the corruption counts, the detection direction on three checkpoints, the 2×2 state main effect, the cross-model mean and the response census. It uses the standard library only, so a reviewer needs no environment beyond Python. Model weights are not redistributed, but each run records the pinned revision and a hash of the weight files before those files are deleted. A permanent archive with a DOI will accompany the camera-ready version.

AI-Use Statement

A large language model was used as a coding and drafting assistant throughout: generating experiment code, analysis scripts and figure code, and drafting this manuscript. All experimental design decisions, the interpretation of results, and the claim boundaries were made and verified by the authors. Every numeric value reported was produced by the released analysis scripts from sealed receipts, not written by a model; the released audits allow any reader to re-derive them independently.

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Table 4: Datasets. The sealed panels are nonce-generated and disjoint; the public corpus validates the certifier only. Experiment 4 rescores Experiment 2’s items rather than generating new ones.

Panel	Classes	Items	Used by
b14	entailed / contradicted	192	Experiment 1
b15	entailed / contradicted	192	Experiment 2
b16	entailed / contradicted / undetermined	192	Experiment 3
b17, b17b	b15 items, three candidates	192	Experiment 4
ProofWriter OWA [Tafjord et al., 2020]	three-way	23,240	certifier validation

Table 5: The four one-factor edges of the 2×2 , and the prespecified diagonal. Post hoc reanalysis of sealed data.

Contrast	Effect	Exact McNemar p	Reported originally
Add answer, no state present	+41.67pp	1.7×10^{-24}	no
Add state, no answer present	+36.98pp	1.5×10^{-19}	secondary
Add answer, state present	+12.50pp	1.2×10^{-7}	no
Add state, answer present	+7.81pp	6.1×10^{-5}	secondary
<i>Diagonal (prespecified primary)</i>	−4.69pp	0.078	primary

Table 6: Signal detection for Experiment 1; hits and false alarms out of 96 each. Apparent chance and zero sensitivity are not the same thing.

Arm	Hits	FA	d'	c
NONE	9	14	−0.26	+1.17
IRRELEVANT	9	9	0.00	+1.29
CONCLUSION_ONLY	80	0	+3.52	+0.81
PROOF_PREFIX	71	0	+3.20	+0.97
FULL	95	0	+4.72	+0.20

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A Supporting Tables and Figures

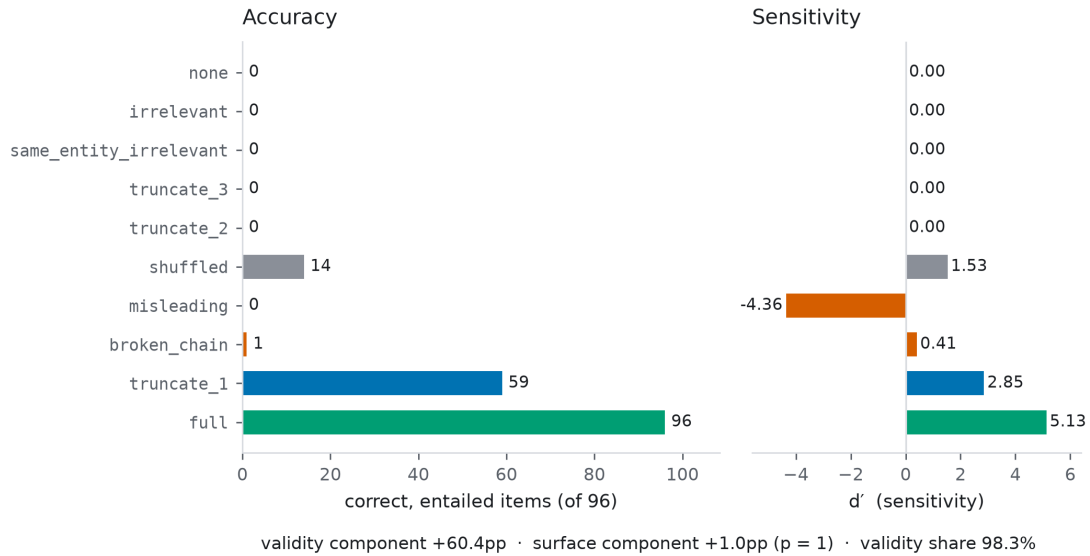


Figure 2: Ten matched arms on one item set, differing only in the solver record. Left: correct on the entailed subset (of 96). Right: sensitivity d' , signed. Destroying validity alone removes almost the whole effect *on this checkpoint*; on gemma-3-4b it does not, see section 5.4. The MISLEADING arm, whose fabricated final rule points the wrong way, is followed to the wrong answer on 189 of 192 items.

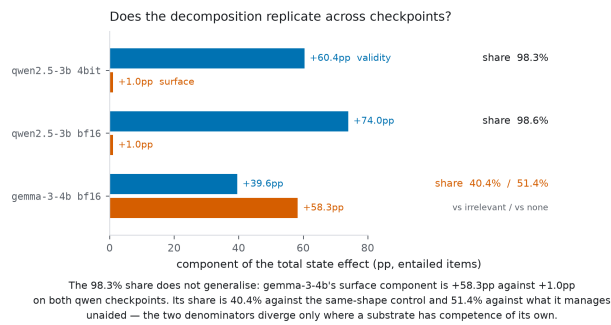


Figure 3: The surface and validity components on three checkpoints. Gemma-3-4B's surface component is +58.3pp against +1.0pp for the two Qwen checkpoints. Bars give the decomposition against the IRRELEVANT control; the share annotation gives both denominators, 40.4% against that control and 51.4% against what the model manages unaided. the two denominators coincide on the Qwen checkpoints, which score 0/96 with no certificate, and diverge only where a substrate has competence of its own.

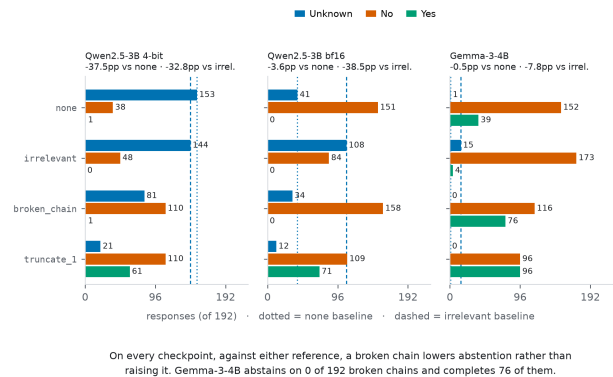


Figure 4: Three-candidate response distribution, three checkpoints, four arms. Dotted and dashed lines mark the NONE and IRRELEVANT abstention baselines, which coincide on the 4-bit checkpoint and diverge at bfloat16. A broken chain lowers abstention against *both* references on every checkpoint — the opposite of the direction validity detection predicts.

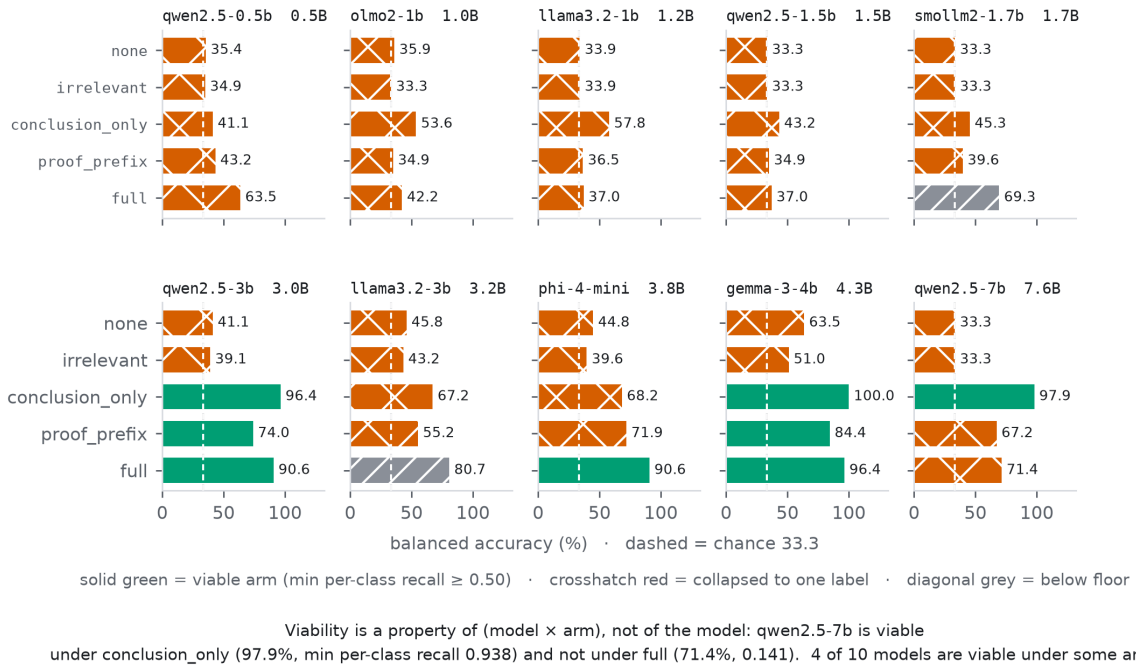


Figure 5: All five arms for all ten models, ordered by parameter count. Chance is 33.3%. Bar colour gives the per-arm viability verdict. Qwen2.5-7B is viable under CONCLUSION_ONLY and not under FULL: viability is a property of (model $\times$ arm).